Week 1–2 Project Report

Data Cleaning & Exploratory Data Analysis (EDA)

I worked on two datasets: the Online Retail Transactions dataset and the Netflix Movies & TV Shows dataset. The goal of this project was to clean raw data, improve its quality, and explore it to uncover meaningful patterns and insights.

The Online Retail dataset contains transaction records from a UK-based online store, including product details, quantities, prices, customer IDs, and countries. The Netflix dataset contains information about movies and TV shows, such as titles, genres, countries, ratings, and release years.

The first stage of the project focused on data cleaning. Using Python and Pandas, I identified and handled missing values, removed duplicate records, standardized column names, corrected data types, and ensured consistency across both datasets. For the Online Retail dataset, I converted the Invoice Date column to a datetime format and created a Revenue column by multiplying Quantity by Unit Price. For the Netflix dataset, I handled missing values in columns such as Director, Cast, and Country while ensuring the data was properly formatted for analysis.

After cleaning the data, I carried out exploratory data analysis to identify trends and patterns. In the Online Retail dataset, I found that the United Kingdom generated the highest revenue, a small number of products accounted for a large share of sales, and revenue showed noticeable fluctuations over time. In the Netflix dataset, movies made up a larger proportion of the content than TV shows, the United States and India were the leading content-producing countries, and Drama-related content appeared most frequently.

One of the most valuable aspects of this project was learning how to troubleshoot common data analysis challenges. I encountered issues such as incorrect file paths, encoding errors, undefined variables, and column naming inconsistencies. Resolving these problems improved my understanding of Python workflows and strengthened my problem-solving skills.

Overall, this project enhanced my ability to clean, validate, and analyze data using Python. It reinforced the importance of data preparation as the foundation of reliable analysis and helped me gain practical experience in handling real-world datasets.